Kai-Hung **Huang**

Senior Software Engineer · Android App & Platform, C++/JAVA/Kotlin & Embedded Systems

Taipei, Taiwan

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Summary

- **10+ Years of Systems Experience:** Proven expertise in embedded Android/Linux, spanning the Framework, native API, and Application layers.
- **6+ Years of AOSP Internals:** Deep specialization in Android Framework customization, including JNI layers and custom System Services.
- **4+ Years of Architecture Design:** Adept at engineering robust mobile applications utilizing MVVM patterns, DI, and reactive data components like LiveData and ViewModel.
- **2+ Years of Engineering Leadership:** Experienced in supervising technical teams to efficiently customize SurfaceFlinger, system-level APIs, and Settings for third-party developer ecosystem integration.
- **BSP & Sensor Integration:** Hands-on technical capability in the Android BSP domain, focusing on sensor driver integration and HAL layers.
- **Collaboration & Mentorship:** Strong cross-functional communicator and team player with a passion for driving technical alignment, mentoring engineers, and solving complex architectural bottlenecks.

Technical Skills

Languages	C/C++, Java, Kotlin, Python, Shell Script
Android OS & Kernel	AOSP, Android Framework (JNI, System Services, SurfaceFlinger), Android NDK, HAL, Linux Systems Programming
Protocols & IPC	Bluetooth (Classic/BLE), Wi-Fi/SoftAP, RTP/RTCP, Unix Domain Sockets, Ashmem
Tools & Debugging	Systrace, Android Studio Profiler, Perfetto, Bugreport Analysis, Git, Audacity

Work Experience

XPERI Corporation (DTS)
Taipei, Taiwan

Sr. Software Engineer — *Aug. 2021 - Present*

- Optimized UI performance by implementing MVVM architecture and RecyclerView caching, eliminating frame drops during data updates.
- Architected custom Android system services for SoftAP, enhancing Wi-Fi network connection stability on embedded hardware platforms.
- Engineered RTP/RTCP streaming protocols for DTS Play-Fi, enabling low-latency multi-channel wireless audio synchronization.
- Implemented a Linux process to process system audio PCM data via Unix Domain Sockets, ensuring low-latency audio on Android TV platforms.

XRSPACE Inc.
Taipei, Taiwan

Lead Programmer — *Mar. 2019 – Aug. 2021*

- Led a team of 4 engineers to deliver core VR features; coordinated cross-functional technical tracks to maximize development efficiency.
- Reduced VR-to-TV streaming latency by customizing SurfaceFlinger and SurfaceView layers to selectively cast VR content via Miracast.
- Developed a secure, low-overhead Bluetooth library for HMD-to-mobile connectivity, implementing custom protocols and data encryption.
- Optimized system stability and eliminated UI thread bottlenecks by pinning critical VR processes using CPU affinity techniques and Systrace analysis.

Sr. Software Engineer — *Sep. 2018 - Mar. 2019*

- Integrated Qualcomm VR and Android Camera NDK to ingest multi-channel feeds (mono, RGB, depth) for 6-DoF tracking algorithms.
- Engineered a zero-copy Ring Buffer utilizing ashmem for high-throughput camera frame transmission to Java/Unity layers.
- Designed high-performance preview windows via ANativeWindow to render 10-bit Raw RGB and YUV-420 formats at stable frame rates.
- Architected the core wireless communication architecture between VR headsets and Android smartphones using Bluetooth LE.

ASUSTeK Computer Inc.
Taipei, Taiwan

Sr. Software Engineer — *Dec. 2015 - Sep. 2018*

- Developed PID line-following and autonomous MCU-integrated docking algorithms for Zenbo/Zenbo Junior.
- Designed and deployed RobotAPI/SDK based on Android system for developer to access proprietary hardware features.
- Integrated battery, facial animation, and motor motion features into RobotFramework via Android System Services.
- Ported capacitive touch Linux drivers and configured GPIO/BIOS across HAL/Framework layers.
- Embedded motion sensors (accelerometer, gyroscope, and compass) via Intel ISH (Sensor Hub).

Education

Tamkang University
New Taipei City, Taiwan

M.Eng., Robotics Engineering — *Mar. 2013 - Aug. 2014*

- GPA: 4.0/4.0; course average: 91.36/100. Completed the program in one academic year.

Tamkang University
New Taipei City, Taiwan

B.Eng., Electrical Engineering — *Mar. 2009 - Aug. 2013*

- GPA: 4.0/4.0; graduated first in class; recipient of Academic Excellence Scholarships for 8 consecutive semesters.